

SELECTED SCIENTIFIC WORK

Research Portfolio

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● COSMIC

● STELLAR

● ATOMIC

THREE RESEARCH PROJECTS

From cosmic mergers to solar plasma
and atomic-scale chemical structure.

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A portfolio organized around three research questions and methods.

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Gravitational waves + cosmology

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Nanoflare heating + solar observations

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Identification of an Unknown Sulfur Compound

XANES + linear-combination fitting

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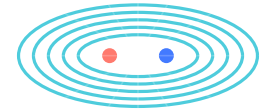
RESEARCH THROUGHLINE

Observe patterns -> compare models -> interpret physical meaning

Across three very different scales, each project turns indirect signals into scientific evidence.

Supermassive Black Hole Mergers: Probing Gravitational Waves and Cosmology

A review of SMBH binaries as astrophysical laboratories and cosmological probes



ABSTRACT

The potential importance of supermassive black holes (SMBHs) in the centers of galaxies to the evolution of galaxies is known. As galaxies merge, their central SMBHs can become binary and eventually coalesce, emitting copious amounts of energy in the form of low-frequency GWs. These mergers provide a unique window into both astrophysics and cosmology - they reveal how galaxies evolve, how black holes transform, and how the large-scale structure of the universe forms. This paper summarizes the current knowledge of SMBH binary formation and evolution, of the theoretical and observational obstacles to modeling their dynamics, and the existing evidence of a gravitational wave background from pulsar timing arrays (PTAs) prospectively. This paper also addresses the prospective event

rate for space-based detectors like the Laser Interferometer Space Antenna (LISA), which can directly detect single SMBH mergers. Multi-messenger astronomy is highlighted, and electromagnetic counterparts, such as AGN flaring or jet activity, can be expected with the GW signal, providing information about the merger environment. Merging SMBHs are likely to be 'standard sirens' when both their electromagnetic and some combination of gravitational wave and electromagnetic signals are observed, allowing for distance measurements and cosmological tests by the independent constraint of the Hubble constant and knowledge of dark energy. Future high-energy observatories should turn SMBH mergers into one of the most powerful probes of the universe.

KEYWORDS

Supermassive black holes / Black hole mergers / Gravitational waves / Pulsar timing arrays

Evaluating Magnetic Reconnection Through the Nanoflare Heating Model

Evidence from modern solar observations



ABSTRACT

The solar corona reaches temperatures exceeding one million Kelvin, far higher than the approximately 5,800 K temperature of the photosphere, giving rise to the long-standing coronal heating problem. This paper reviews modern observational evidence related to the nanoflare heating model and discusses the extent to which it supports magnetic reconnection as a mechanism for coronal heating. A qualitative literature review was conducted using three representative studies that provide complementary perspectives: theoretical predictions of nanoflare heating, observational evidence from solar active regions, and the thermal evolution of individual coronal loops. The studies were compared in terms of observed heating signatures, consistency with theoretical predictions, and their main limitations. The reviewed evidence shows several behaviors consistent with impulsive nanoflare heating, including temperature-dependent emission sequences, heating and cooling patterns in coronal loops, and

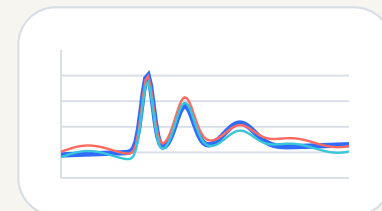
signatures associated with small-scale episodic energy release. These characteristics are observed across different spatial scales and generally agree with predictions of nanoflare-based models. However, the evidence remains indirect because individual nanoflares and their associated magnetic reconnection events cannot yet be directly resolved by current instruments. In addition, observational uncertainties and differences between modeled and observed loop behavior limit the strength of the conclusions. Overall, modern solar observations support nanoflares as an important potential contributor to coronal heating, but they do not establish magnetic reconnection through nanoflares as the sole primary heating mechanism. Further improvements in observational resolution and theoretical modeling are required to better determine the role of nanoflares in coronal heating.

KEYWORDS

Coronal heating / Nanoflares / Magnetic reconnection / Solar corona / Coronal loops

Identification of an Unknown Sulfur Compound Using XANES and Linear-Combination Fitting

Using sulfur K-edge spectra to infer chemical environment



ABSTRACT

To identify the chemical environment, oxidation state, and bonding of an element - factors that directly affect how a material behaves and reacts - simply confirming the presence of an element is insufficient, because it cannot reveal how the element is bonded or what form it takes; XANES addresses this issue, as spectral features near the absorption edge change with the electronic and chemical environment of the element, and these changes can be studied through the position and shape of the features. For sulfur, sulfur K-edge XANES can distinguish different sulfur species even when they all contain sulfur. In response to the challenge of identifying an unknown sulfur-containing compound, this paper proposes a method based on sulfur K-edge XANES, which is sensitive to the electronic structure and local chemical environment of sulfur and can

therefore be used to distinguish different sulfur compounds. This study combines experimental XANES data processing with reference-spectrum fitting: after background subtraction and normalization, the unknown spectrum was compared with the reference spectra of dibenzothiophene (DBT), dibenzyl sulfide, sulfoxide, and sulfone, and linear-combination fitting was performed using Excel Solver to determine the reference combination that best reproduced the unknown spectrum. The results show that the unknown sample is most consistent with dibenzothiophene (DBT) among the reference compounds tested. This study demonstrates that sulfur K-edge XANES can be used to identify chemicals in an unknown sample by their chemical environment.

KEYWORDS

XANES / Sulfur K-edge / Dibenzothiophene / Linear-combination fitting